

CS201 Midterm 1 Part A

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I. QUESTION (ALGORITHM ANALYSIS) (20 POINTS)

```

1 int algorithm1(int N){
2     int i , j , k, sum = 0;
3     for (i = 1; i <= N - i; i++)
4         sum++;
5     for (i = 1; i <= N; i++)
6         for (j = 1; j <= N; j *= 2)
7             sum++;
8     for (i = 0; i <= N; i++)
9         for (j = i + 1; j <= N - i; j++)
10            sum++;
11     return sum;
12 }
```

What does the function algorithm1 return in terms of N? Assume that N is a power of 2.

II. QUESTION (ALGORITHM ANALYSIS) (20 POINTS)

```

1 int algorithm2(int N){
2     if (N == 1)
3         return 0;
4     int sum = 0;
5     for (int i = N; i > 0; i--)
6         for (int j = 0; j < N; j++)
7             for (int k = 0; k < N; k++)
8                 sum++;
9     for (int i = 0; i < 8; i++)
10        for (int j = 0; j < 2; j++)
11            sum += algorithm2(N / 4);
12     return sum;
13 }
```

What is the time complexity of algorithm2? You must construct the recurrence equation and solve it without using the master theorem.